

EA4EOZ, an amateur radio electronic enthusiast

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Friday, September 8, 2017

Repairing the OM345 hybrid module



Recently I had to repair a Marconi 2022E signal generator. It's a nice unit, relatively small, and well suited for amateur use. Talking about repairing the old Marconi instrumentation is talking about the Phillips OM345 hybrid modules. Why? Because they has a really high failure rate and old Marconi equipment make extensive use of these modules.

The module has been a pain to replace until recently. Thanks to the investigations of some very talented people the hybrid has been teared down and reversed engineered. This enable you to repair these modules without too much effort.

You can find these information in Internet but scattered in some forum threads and personal pages but there is one that stands out from all the others:

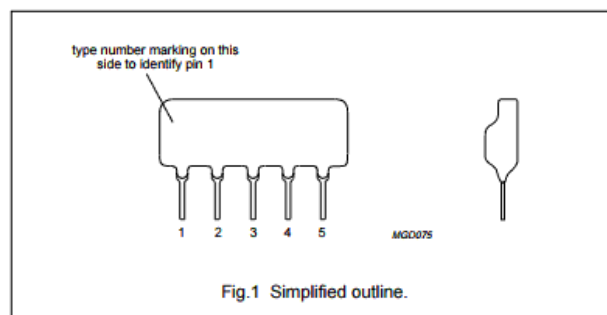
<https://www.vintage-radio.net/forum/showthread.php?t=100840>

so I decided to summarize all the information in a single page for convenient access.

The OM345 hybrid module

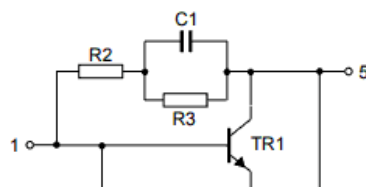
The module is an small piece of ceramic covered in resin with 5 pins. This module was very common in mast head TV preamplifiers in the 80s. In fact, the working frequency range according to the datasheet is 40 to 860 MHz and its input and output impedance is 75 ohms. Its gain is about 12dB with a noise figure of 5.5 dB. As you can see nothing special for today's standards.

PIN	DESCRIPTION
1	input
2	common
3	common
4	supply (+)
5	output



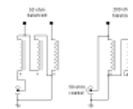
OM345 pinout

In Marconi equipment these modules are used as block gain devices. Wherever it was needed to amplify a signal, a OM345 module was used. You can see it like an ancient MMIC.

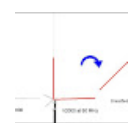


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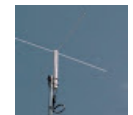
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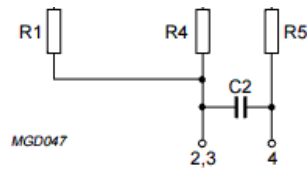
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OM345 simplified diagram

This is the simplified diagram you can find in the datasheet. The interesting area is R2-R3-C1. R2 is a 470 ohms resistor, R3 is a 3900 ohms resistor and C1 is a 1nF capacitor.

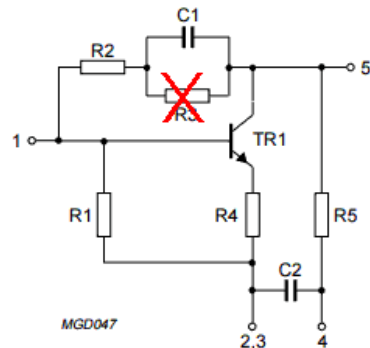
Testing the module

The quickest way to test the module is to measure voltages at input and output pins. First make sure the module is powered: In some Marconi equipment they are powered on / off by relays according the instrument settings. This is as easy as check voltage at pin 4. If you see 12 volts, it is powered on.

Now measure voltages at input (pin 1) and output (pin 5). Pin 1 must be around 0.8 volts and pin 5 should has roughly half supply voltage, around 6 - 7 volts. If you have measure these voltages the module is almost for sure ok. In the most common failure mode, you will see zero volts at input pin, and 12 volts on output pin. If you see these voltages, your module is faulty, but probably you can bring it back to life. Keep reading!

The failure

The most common failure of this module is not the transistor as you can think. The most common failure of this module is R3. It is a very small printed resistor with a value of 3900 ohms.



The most common failure in OM345 hybrid modules

It is not clear if the failure is for excess of dissipation or because mechanical stress due to thermal cycles. Anyway, the resistor gets open, the transistor loses its polarization, so the base goes to zero volts. With no base current the transistor goes to cut-off, so collector goes to the supply voltage, 12 volts.

Desoldering an OM345 hybrid module

A small warning to you. Desoldering a OM345 module is a very delicate procedure. Although it has only 5 pins, the pins are soldered to the small ceramic substrate, and once you have put enough heat on a pin to melt the tin, the only thing keeping it pin in place is the resin. If you force or twist the module while heating, the resin can break, and then the pin will fall. It is not a catastrophe, it can be easily solved, but it is not pretty.



spectrum ana...

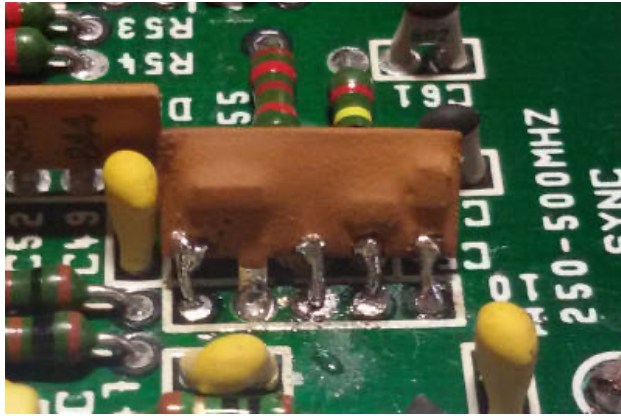
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About Me



Miguel A. Vallejo,
EA4EOZ

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Restoring pins on OM345 module

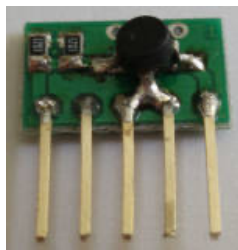
Your options

Once you have identified a faulty module you have some options:

Replace it with another OM345. This is the most obvious option. The problem is OM345 are really hard to find today, and if you are lucky enough to find one, it will cost you a small fortune. And I bet you you will find more than one damaged hybrid in your Marconi equipment, so it will be really expensive option.

Replace it with a OM2045 module. The OM2045 module is a direct replacement for OM345. Same specifications with a bit lower noise figure (3.5 dB vs 5.5 dB). The problem is the same as for OM345: it is really hard to find today, and if you are lucky enough to find one, it will cost you a small fortune.

Replace it with a MMIC. This was the most common option. Because the OM345 is used just as a gain block, it is easily replaceable by a MMIC. Pin 1 goes to MMIC input, pin 2 and 3 to MMIC ground, pin 4 through a resistor to MMIC output, who is also connected to pin 5. Many people sell small PCBs with all components and pins, ready to be used. Just make a search in Internet and you will find them really soon.

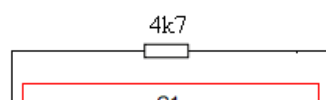


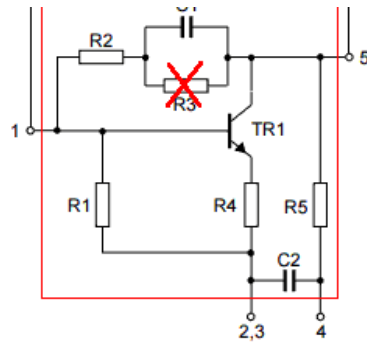
But of course you can make your own replacements. Just make sure you select an MMIC with at least 12 dB of gain in the range 0-1 GHz. Common choices are MAR-1, MAR-2, MAR-3, MAR-5, MAR-6, MAR-7, ERA-1, ERA-2, ERA-4, MSA-0735, and many others.

Keep in mind most of these MMICs have an P1dB / IP3 points lower than OM345 has. They will work nicely in some places, but they will degrade in some degree the signal in other places. If the module is the one that feeds a prescaler (Marconi 2022) or feed the frequency counter (Marconi 2955) it will work just fine, but if the generated signal goes through them, it could have a higher distortion and/or content in harmonics. Always test the output signal after a module replacement.

Thanks to the work of G0HZU, there are two more options to solve the problem of a faulty OM345 hybrid module:

Place a 4.7k resistor between input and output. In the most common failure mode R3 gets broken and becomes open circuit, so the transistor loses its DC polarization.

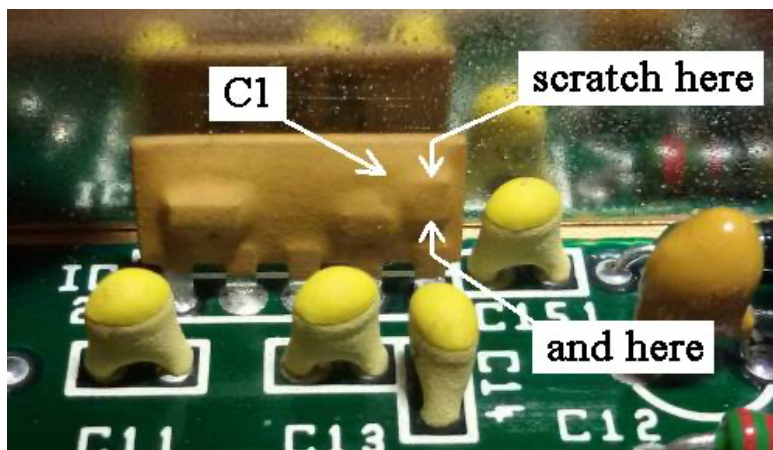




Restoring DC polarization

Placing a 4.7k resistor between pins 1 and 5 restores the DC polarization and the hybrid comes back to life. This works because the RF negative feedback is still intact in the module (C1 and R2). Placing the DC feedback outside the module produces small variations in specifications but nothing serious. It will work just fine.

Just replace R3: R3 is a small film resistor printed in the ceramic substrate so it is not easily replaceable. But you have access to its terminals in C1, and C1 is quite easy to access in the module: it is the rightmost bump in the back of the module.



Accessing C1 in OM345 hybrid module

Just scratch with a sharp object either side of C1 to expose its pads. Then solder a 3.9k resistor over it, in parallel. A 0603 SMD resistor fits perfectly over C1. This is by far the best option to repair a faulty OM345 hybrid module and the recommended one.

Conclusion

I think G0HZU's work marked a turning point. Before, replacing these modules was expensive and MMIC substitution, although easy was not immediate. After his work, everything become easier, usually as easy as place a resistor, over a capacitor or over the input/output pins.

In my case I tried to desolder the modules to add the 3.9k resistor over C1, but when I tried to desolder the first one 4 of the 5 pins fall. So I decided the 4.7k resistor trick was also ok :-)

I restored the missing pins and placed 1/4 watt 4.7k resistors between input/output pins over the four faulty modules I found in the Marconi 2022E. The signal generator came back to life and everything seemed to work just fine.

So my advice is if you find a faulty OM345 with no voltage at input pin and 12 volts on output pin, and you can desolder it with guarantee, try the 3.9k resistor over C1. Otherwise just place a 4.7k resistor between input and output pins. Probably that is all you need to bring the module back to life.

I want to thank G0HZU for his work and share it with everyone. Thanks to him I have a nice working signal generator.

Miguel A. Vallejo, EA4EOZ

Publicado por Miguel A. Vallejo, EA4EOZ en [6:37 PM](#)

Etiquetas: [Electronics](#)

22 comments:



klarkkk December 26, 2017 at 12:08 PM

it works! Thank you, works at first shot.

[Reply](#)

Anonymous October 15, 2018 at 12:38 AM

Reparación por el método de la resistencia smd de 3k9 100% ok en mi 2022C, en dos de los OM345. Muchas gracias Miguel!!

[Reply](#)

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Anonymous October 16, 2018 at 10:32 PM

Se me olvidó decir que soy Javier Muriedas, EA1AWY



Miguel A. Vallejo, EA4EOZ January 19, 2019 at 11:19 PM

Me alegro de que la información te haya servido. ¡Saludos!

[Reply](#)



Unknown January 19, 2019 at 3:19 AM

Great writeup on OM345 modules, fixes and replacement options. I picked up a Marconi 2022 recently that had problems with the 1dB step attenuator above 250 MHz and frequency error below 64 MHz. With the help of the service manual descriptions and schematics and your detailed advice here both problems were fixed at the cost of a couple of 4.7K resistors using the OM345 quick fix option. The generator is well within specs with those two quick repairs and it took roughly half an hour of reading the circuit descriptions and schematics and about fifteen minutes with the unit opened up to trace down the problems and tack solder in the fixes. It probably took longer to run through the full check out across all sub bands and amplitudes than it did to troubleshoot the problems but it wouldn't have been nearly that easy without the tremendous info you supply here. Thanks!

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Miguel A. Vallejo, EA4EOZ January 19, 2019 at 11:22 PM

I'm glad to read this info has been useful to you. 73!

[Reply](#)

G4NCE May 10, 2019 at 5:58 PM

Just fixed a non-variable sig gen fault on my 2955A thanks to your article Miguel. AA4 board IC5 faulty. Very well written and very useful. Thanks. Martin

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Miguel A. Vallejo, EA4EOZ July 25, 2019 at 12:01 AM



Thanks for sharing your experience. 73!

[Reply](#)



Unknown November 24, 2019 at 10:12 AM

Thank you dear OM, your info gave me the possibility to repair un SMG generator from RS
Simply the solution to save the OM345 was the 4.7K resistor
73
ON7MB

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Miguel A. Vallejo, EA4EOZ January 12, 2021 at 11:05 AM

Happy to help you. 73!

[Reply](#)



Termite May 31, 2020 at 5:53 PM

but what if you have errors 25 and 27 but you have signal on all four OM345 and you have correct voltages also..
can it still be the modules ?
would i be replacing a good module for a copy module and it not work ?

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Miguel A. Vallejo, EA4EOZ May 31, 2020 at 10:57 PM

The OM345 modules are the most probable cause of failure, but not the only one. There are also many several decades old components that can fail as well, for example, electrolytic capacitors.

Good luck!



Termite June 1, 2020 at 8:52 PM

thank you, i have put 4.7k resistor on each and all OM345 still have fault.
so i can guess the OM345 are ok ??
so i am look at maybe bad capacitors or other part as well ??



Miguel A. Vallejo, EA4EOZ June 2, 2020 at 1:29 AM

Don't put the resistor is voltages are fine: about 6-7 volts at output and zero at input, otherwise you can damage a working module.

It seems your modules are ok, so your problem is elsewhere. Good luck!



Termite April 1, 2022 at 5:06 PM

since this post my 2955 started to work all good until recently when i suspect OM345 have both failed (IC1 and IC2)..
i have the 4.7k resistor over 1 to 5 pins and is working good.
i have just bought the 3k9 SMD so that is my next job, already cleaned the coating off ready..
so hopefully this will be a permanent fix..

[Reply](#)

Gabriele IK6QNE July 29, 2020 at 10:49 PM

Your page was very interesting and helpfull to solve my problem on 2955a module OM345. Thanks a lot. best 73 Gabriele

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Miguel A. Vallejo, EA4EOZ

July 30, 2020 at 2:15 PM

Thank you

[Reply](#)

Anonymous [February 17, 2022 at 10:49 AM](#)

Top Man.. did just repair my MARconi 2022E, with a 1 Eurocent Resistor..

[Reply](#)

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Miguel A. Vallejo, EA4EOZ

February 17, 2022 at 2:51 PM

Congratulations!

[Reply](#)

Juan M Chazarra EA5RS [September 25, 2022 at 11:47 AM](#)

Gracias Miguel A. He reparado mi generador R&S SMG en dos ocasiones de esta forma. Los dos OM345 que lleva han fallado con unos meses de intervalo.

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Miguel A. Vallejo, EA4EOZ

September 25, 2022 at 2:36 PM

Genial!

[Reply](#)

Anonymous [December 29, 2023 at 6:14 PM](#)

Fantastic. I superb solution. Many thanks. de M0MGA

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